2-Acre Moringa Drip Irrigation

Stage-by-Stage Engineering Execution Briefing

Author: ArtCraft Suite (DeckCraft & CADCraft)

Project Scope: Solar Borewell Head, 2" Mainline, 8 Risers, 56 Laterals

Stage 1: Borewell Head & 2" Mainline Routing

- • Solar Borewell Head Unit at (590, 20): 5 HP pump delivery with 2" NRV, Gate Valve, and 2" T-Type Filter.
- • Northward Mainline (80'-0"): 2" PVC Class 2 from Borewell to Corner Tee (590, 100).
- • Westward Mainline (90'-0"): From Corner Tee to Tee A (500, 100) — Plot A Center Entry.
- • Inter-Plot Mainline (215'-0"): Runs West from Tee A under Plot A & 25ft buffer road to Tee B (285, 100).
- • Trench Depth: Minimum 2'-6" deep with 2" sand bed and 24-hr solvent cement cure.

Stage 2: Plot A Risers & 24 Inline Trenches (PKM-1)

- • Center Mainline: Runs North 145'-0" from Tee A to End Cap (500, 245).
- • 4 Overhead Riser Stations: A1 (30'), A2 (60'), A3 (90'), A4 (120') with 4ft double-tee manifold.
- • Subline Network: 1.5" (50mm) PVC running 15ft South and 15ft North from each station (8 arms = 120 ft total).
- • 24 Active Trenches: Spaced 5ft apart with 50x16x2-way take-off collars (V-connectors).
- • Laterals: 16x4x40 Inline Pipe (2,926m total) running 195ft East and 195ft West for 3,000 PKM-1 trees.

Stage 3: Plot B Risers & 32 Online Rows (Bhagya)

- • East-Boundary Mainline: Runs North 240'-0" from Tee B along Plot B edge to End Cap (285, 340).
- • 4 Overhead Riser Stations: B1 (24'), B2 (88'), B3 (152'), B4 (216') with 2"x1.25" reducers.
- • Subline Network: 1.25" (40mm) PVC running 32ft South and 32ft North from each station (8 arms = 256 ft total).
- • 32 Active Rows: Spaced 8ft apart with 40x16x1-way TF collars pointing West only.
- • Plain Pipe & Punching: 16mm Plain Pipe (1,610m) with 768 Tunning Drippers punched every 7ft (750 trees).

Stage 4: Systematic Flushing & Pressure Testing

- • Phase A — Mainline Flush: Open 2" end caps at (500, 245) and (285, 340) for 5 mins to clear construction sand.
- • Phase B — Subline Flush: Close mainline; open all 16 flush valves (50mm & 40mm) until discharge runs clear.
- • Phase C — Lateral Flush: Remove all 80 lateral endcaps (48 in Plot A + 32 in Plot B) for 3 mins.
- • Phase D — Pressure Test: Hold 2.5 kg/cm² test pressure for 60 mins; verify zero joint weeping and uniform emitter discharge.

Stage 5: Commissioning, Flow Audit & Handover

- • Emitter Uniformity Audit: Catch-can sampling across 20 points per zone; target Christiansen Uniformity > 92%.
- • Daily Irrigation Regimen: Plot A (3,000 plants @ 4 LPH = 12,000 LPH); Plot B (750 plants @ 35 LPH = 26,250 LPH peak).
- • Zone Rotation: Rotate irrigation between Zone 2 and Zone 3 to balance solar pump curve throughout peak sunlight hours.
- • Maintenance Protocol: Daily filter purge, weekly disc washing, monthly flush valve blow-out.